

# Computer Applications in Management

Complete handbook covering computer hardware, operating systems, data processing, networks — with Malayalam support, interactive calculators, diagrams, and exam preparation.

**PROGRAMME****Computer Application & Business Management****COURSE CODE****1254****CREDITS****4****ESE MARKS****100****COURSE TITLE****Computer Applications in Management****TEACHING SCHEME****3:0:2:0 (L:T:P:R)****CIE MARKS****40****SEMESTER****I**

## Source Declaration

This handbook is based on the official Kerala Polytechnic Revision 2026 syllabus for Course **1254 — Computer Applications in Management**.

 **Source integrity:** The syllabus PDF was used as the authoritative source. No external materials were consulted. The handbook follows the official course structure, learning outcomes, and assessment scheme exactly as specified.

 **Verified inconsistencies:** The PDF contains some minor formatting inconsistencies in the teaching scheme table (the header "Self-learning (SS)/Week" appears with a colon but no value in the row) and the practical assessment table has some column alignment variations. These have been interpreted according to the most consistent reading across the document. The handbook presents the information as it is most coherently understood from the official source.

REV2026

SOURCE: SITTR

COURSE 1254

## Course Dashboard

Key statistics and structure of Course 1254 — Computer Applications in Management.

**4****Modules**

Covering computer basics, OS, data processing, and networks

**60****Contact Hours**

Total instructional hours across theory and practical

**4****Credits**

Practicum course with 3:0:2:0 teaching scheme

140

Total Marks

40 CIE + 100 ESE

| Teaching Scheme/Week | Self-learning/Week | Total Hours/Semester | Credits | CIE |    | ESE |    | Total |
|----------------------|--------------------|----------------------|---------|-----|----|-----|----|-------|
| 3:0:2:0              | 5.5                | 60                   | 4       | 20  | 20 | 60  | 40 | 140   |

Teaching and Assessment Scheme — official structure



Pre-requisites

Basic computer knowledge at the Secondary Education level.



Course Type

Practicum — hands-on laboratory work with theory components.

LEARNING PATH

How to Use This Handbook

Follow this four-step roadmap to get the most out of your study.

1

Understand

Read each module's theory, definitions, and concepts. Use the Malayalam notes for clarity.

2

Visualise

Study the SVG diagrams and network topology illustrations. Watch the animated generators.

3

Apply

Work through the calculators, practical procedures, and self-learning activities.

4

Revise

Use the Q&A, model paper, revision cards, and glossary to cement your knowledge.

WHAT YOU WILL LEARN

Course Outcomes

Official course objectives and student learning outcomes mapped to Bloom's Taxonomy.

### Course Objectives

1. Provide students with theoretical and practical knowledge of computer hardware and software basics.
2. Lay foundation in operating systems, data processing, data communication, networks, assembling, and software installation.
3. Explain data communication concepts and identify network components in digital environments.
4. Identify different types of computer networks and compare topologies for organisational requirements.

### CO-PO Mapping

Course Outcomes mapped to Program Outcomes (PO1-PO11). Correlation: 1=Low, 2=Medium, 3=High.

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 3   | 2   | -   | -   | 2   | -   | -   | -   | -   | -    | -    |
| CO2 | 3   | 2   | 2   | -   | 3   | -   | -   | -   | -   | -    | -    |
| CO3 | 3   | 2   | -   | 2   | 2   | -   | -   | -   | -   | -    | -    |
| CO4 | 3   | 2   | -   | 2   | 2   | -   | -   | -   | -   | -    | -    |

| CO  | Course Outcome                                                                                                                         | Bloom's Level |
|-----|----------------------------------------------------------------------------------------------------------------------------------------|---------------|
| CO1 | Explain the basics of computer system and identify the components                                                                      | Apply         |
| CO2 | Understand the types and functions of the operating systems and apply the knowledge to install various system and application software | Apply         |
| CO3 | Define Data processing, identify the communication elements and types of data transmission media                                       | Apply         |
| CO4 | Identify and illustrate various computer networks and topologies                                                                       | Apply         |

*Course Outcomes with Bloom's Taxonomy Level*

### SYLLABUS MAP

## Curriculum Coverage

Every module, topic, and subtopic from the official syllabus is covered in this handbook.

| Module | Title                     | Major Topics                                                                             | L  | P | CO  |
|--------|---------------------------|------------------------------------------------------------------------------------------|----|---|-----|
| I      | Basics of computer system | Introduction, Characteristics, Generations, Components, Memory, Hardware/Software, Types | 12 | 0 | CO1 |

| Module | Title                                                              | Major Topics                                                                                                                          | L  | P | CO  |
|--------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----|---|-----|
|        |                                                                    | of Software, Languages, Translators, Algorithm, Flowchart                                                                             |    |   |     |
| I      | Components identification & assembling                             | Motherboard, memory, HDD, CMOS, SMPS, power connector, keyboard connector, Desktop & Laptop assembling, checking system configuration | 0  | 6 | CO1 |
| II     | Operating system                                                   | Definition, Functions, Layers, Types, Techniques, Popular OS, Mobile OS                                                               | 11 | 0 | CO2 |
| II     | Installation of various OS and application software                | Windows, Linux, Application software installation                                                                                     | 0  | 8 | CO2 |
| III    | Data processing and Communication                                  | Data Processing, File Management, DBMS, Communication elements, Modes, Transmission media, Modems, Wireless                           | 11 | 0 | CO3 |
| III    | Data processing & demonstration of cables                          | Data processing, file types, guided media, modems, wifi, UTP crimping                                                                 | 0  | 8 | CO3 |
| IV     | Computer Networks and Topologies                                   | Networks definition, advantages, types, topologies, components, Internet basics, Cloud Storage                                        | 11 | 0 | CO4 |
| IV     | Basic network troubleshooting and demonstrating network components | Troubleshooting, printer connection, network connectivity, search engines                                                             | 0  | 8 | CO4 |

*Module-wise coverage with instructional hours and mapped CO*

## HOW MODULES CONNECT

### Concept Roadmap

Each module builds on the previous one, creating a coherent learning journey from hardware to networks.

#### **Module I**

**Computer Basics** — Hardware, software, memory, languages, and assembling.

#### **Module II**

**Operating Systems** — OS functions, types, installation, and mobile OS.

#### **Module III**

**Data Processing & Communication** — File management, DBMS, transmission media, modems.

#### **Module IV**

**Computer Networks** — Topologies, components, internet basics, cloud storage.

 **Learning flow:** Start with Module I to understand the machine itself, then Module II to see how software controls it, Module III to handle data and communication, and Module IV to connect everything in a network.

## MODULE I

# Basics of Computer System & Components Identification

Understanding the fundamental building blocks of a computer — from hardware components to software, memory, languages, and assembling.

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Theory: 12 hours

Practical: 6 hours

CO: CO1

Bloom: Apply

# Introduction to Computers — Characteristics & Generations

**What is a Computer?** A computer is an electronic device that processes data, following a set of instructions (a program) to produce meaningful information. It accepts input, processes it, stores it, and produces output.

## Characteristics of Computers

- **Speed:** Computers can process millions of instructions per second. Measured in MIPS (Million Instructions Per Second) or clock speed (GHz).
- **Accuracy:** Computers produce error-free results as long as the input and instructions are correct. Errors are typically due to human mistakes (garbage in, garbage out).
- **Storage:** Computers can store vast amounts of data permanently (secondary storage) and temporarily (primary memory).
- **Versatility:** Computers can perform a wide variety of tasks — from calculations to graphics, gaming, and data analysis.
- **Diligence:** Computers can work for hours without fatigue, performing repetitive tasks with consistent accuracy.
- **Automation:** Once programmed, computers can operate automatically without human intervention.

## Generations of Computers

| Generation               | Technology                                 | Key Features                                                         | Examples                      |
|--------------------------|--------------------------------------------|----------------------------------------------------------------------|-------------------------------|
| First (1940s-1956)       | Vacuum Tubes                               | Large, expensive, consumed huge power, magnetic drums for memory     | ENIAC, UNIVAC                 |
| Second (1956-1963)       | Transistors                                | Smaller, more reliable, less power, magnetic core memory             | IBM 1401, CDC 1604            |
| Third (1964-1971)        | Integrated Circuits (ICs)                  | Multiple transistors on a single chip, keyboards/monitors introduced | IBM 360, PDP-8                |
| Fourth (1971-Present)    | Microprocessors                            | Thousands of ICs on one chip, personal computers, GUI, networking    | IBM PC, Apple Macintosh       |
| Fifth (Present & Beyond) | Artificial Intelligence, Quantum Computing | Parallel processing, AI, natural language, quantum bits              | AI systems, Quantum computers |

Computer Generations — key characteristics

**Malayalam support:** മലയാളത്തിൽ പരസ്യങ്ങൾ ചെയ്യാൻ ആഗ്രഹിക്കുന്നവർക്ക് മലയാളത്തിൽ പരസ്യം ചെയ്യാൻ സാധിക്കും. പരസ്യം, വാർത്ത, വാർത്താ പരസ്യം, വാർത്താ പരസ്യം എന്നിവയ്ക്ക് മലയാളത്തിൽ പരസ്യം ചെയ്യാൻ സാധിക്കും. പരസ്യം ചെയ്യാൻ സാധിക്കും പരസ്യം ചെയ്യാൻ സാധിക്കും, IC-ൽ, പരസ്യം ചെയ്യാൻ സാധിക്കും, AI പരസ്യം ചെയ്യാൻ സാധിക്കും.

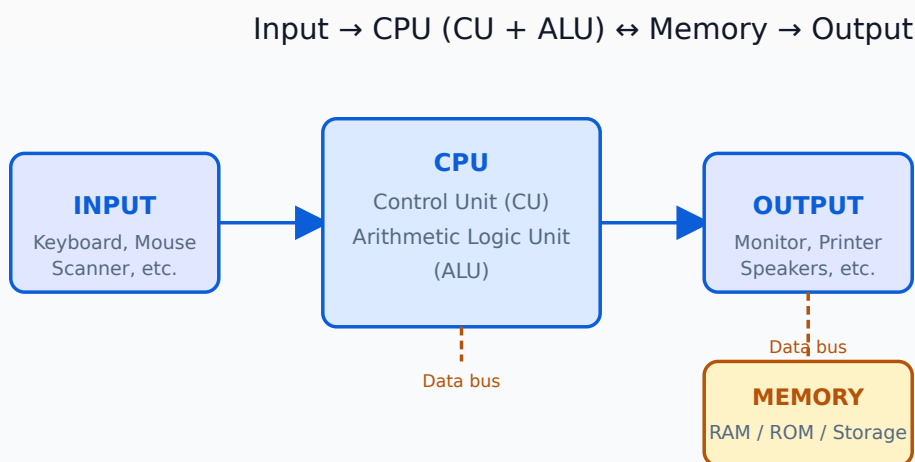


## Components of Computer Systems

A computer system consists of **hardware** (physical components) and **software** (programs and data). The hardware components work together in a coordinated manner.

### Major Components

- **Input Unit:** Keyboard, mouse, scanner, microphone — devices that feed data into the computer.
- **Central Processing Unit (CPU):** The "brain" of the computer. Contains the Control Unit (CU) and Arithmetic Logic Unit (ALU).
- **Memory Unit:** Stores data and instructions. Includes primary memory (RAM, ROM) and secondary storage (hard drives, SSDs).
- **Output Unit:** Monitor, printer, speakers — devices that display or present processed data.
- **Motherboard:** The main circuit board that connects all components.



*Block diagram showing the flow of data: Input → CPU (CU + ALU) ↔ Memory → Output*

**Malayalam support:** കമ്പ്യൂട്ടർ സിസ്റ്റത്തിന്റെ പ്രധാന ഘടകങ്ങൾ:   
ഘടകങ്ങൾ (ഇൻപുട്ട്, ഔട്ട്പുട്ട്), CPU (കേന്ദ്ര പ്രോസസ്സിംഗ് യൂണിറ്റ്), ഓപ്പറേറ്റിംഗ് (RAM, ROM,   
സ്റ്റോറേജ്), ഔട്ട്പുട്ട് ഘടകങ്ങൾ (മോണിറ്റർ, പ്രിന്റർ). കമ്പ്യൂട്ടർ   
ഘടകങ്ങൾ ഒരു കമ്പ്യൂട്ടർ സിസ്റ്റത്തിൽ ഉൾപ്പെടുന്നു.

## Computer Memory — Types and Hierarchy

Computer memory is used to store data and instructions. It is organised in a hierarchy from fastest (and most expensive) to slowest (and cheapest).

### Types of Memory

- **Primary Memory (Main Memory):**

- **RAM (Random Access Memory):** Volatile — loses data when power is off. Used for running programs and current data.
- **ROM (Read Only Memory):** Non-volatile — retains data without power. Contains firmware (BIOS).
- **Cache Memory:** Very fast, small memory between CPU and RAM. Stores frequently accessed data for speed.

- **Secondary Memory (Storage):**

- **Hard Disk Drive (HDD):** Magnetic storage, large capacity, slower access.
- **Solid State Drive (SSD):** Flash-based, faster than HDD, no moving parts.
- **Optical Media:** CD, DVD, Blu-ray — read/write using lasers.
- **Removable Media:** USB flash drives, memory cards.

### Memory Hierarchy (Speed & Cost)

| Memory Type              | Speed       | Volatility   | Cost per GB | Typical Size |
|--------------------------|-------------|--------------|-------------|--------------|
| Register (CPU internal)  | Fastest     | Volatile     | Highest     | Few bytes    |
| Cache (L1/L2/L3)         | Very fast   | Volatile     | Very high   | MB           |
| RAM (DRAM/SRAM)          | Fast        | Volatile     | High        | GB           |
| SSD (Solid State Drive)  | Medium-fast | Non-volatile | Medium      | GB-TB        |
| HDD (Hard Disk Drive)    | Slow        | Non-volatile | Low         | TB           |
| Optical (CD/DVD/Blu-ray) | Very slow   | Non-volatile | Lowest      | GB           |

*Memory hierarchy — from fastest to slowest*

 **Common mistake:** Students often confuse RAM and ROM. Remember: **RAM** is volatile and can be written to; **ROM** is non-volatile and is typically read-only (though some types like EEPROM can be written with special procedures).



# Computer Languages & Translators

Computers understand instructions in binary (0s and 1s). Different levels of programming languages bridge the gap between human-readable code and machine-executable instructions.

## Levels of Programming Languages

- **Machine Language (1GL):** Binary code (0s and 1s). The only language directly understood by the CPU. Fast but difficult for humans to write.
- **Assembly Language (2GL):** Uses mnemonics (e.g., MOV, ADD) instead of binary. Needs an assembler to convert to machine code.
- **High-Level Language (3GL):** Uses English-like syntax (e.g., C, C++, Python, Java). Needs a compiler or interpreter to translate.
- **Fourth-Generation Language (4GL):** Designed for specific applications (e.g., SQL for databases).
- **Fifth-Generation Language (5GL):** Used for artificial intelligence and problem-solving (e.g., Prolog).

## Translators

- **Compiler:** Translates the entire program at once, producing an executable file. Fast execution, but errors are found after compilation. Example: GCC (C compiler).
- **Interpreter:** Translates and executes one line at a time. Slower execution, but errors are found immediately. Example: Python interpreter.
- **Assembler:** Translates assembly language into machine code.

| Feature         | Compiler              | Interpreter        |
|-----------------|-----------------------|--------------------|
| Translation     | Whole program at once | Line by line       |
| Execution Speed | Faster                | Slower             |
| Error Detection | After compilation     | During execution   |
| Output          | Executable file       | No executable file |
| Example         | C, C++                | Python, JavaScript |

Compiler vs Interpreter

**Malayalam support:** മലയാളത്തിൽ പ്രോഗ്രാമിംഗ് ഭാഷകൾ: ബൈനറി (0/1), അസംബ്ലി (mnemonics), ഉയർന്ന-തല (English-like). ഫോർത്തി-തല പ്രോഗ്രാമിംഗ് ഭാഷകൾ: ക്വെറി ഭാഷകൾ (SQL), ഫോർത്തി-തല പ്രോഗ്രാമിംഗ് ഭാഷകൾ (Prolog).

## Algorithm & Flowchart

**Algorithm:** A step-by-step procedure to solve a problem. It is written in plain English or pseudo-code. An algorithm must be finite, unambiguous, and effective.

**Flowchart:** A graphical representation of an algorithm using standard symbols. It visualises the flow of control and data.

### Common Flowchart Symbols

- **Oval (Terminal):** Start / End
- **Parallelogram (Input/Output):** Read / Print
- **Rectangle (Process):** Calculation / Assignment
- **Diamond (Decision):** Condition / Branching
- **Arrows (Flow lines):** Direction of flow

#### **Example: Algorithm to find the sum of two numbers**

**Step 1:** Start

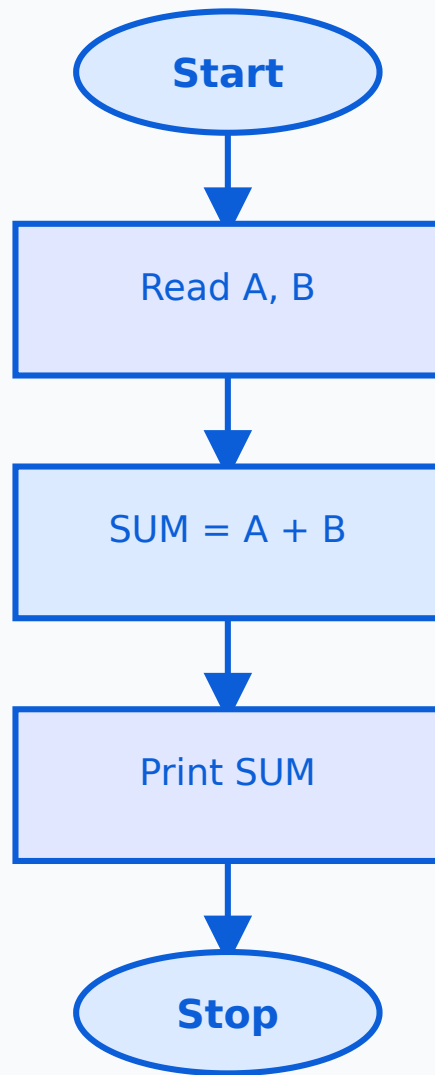
**Step 2:** Read two numbers A and B

**Step 3:** Calculate  $SUM = A + B$

**Step 4:** Print SUM

**Step 5:** Stop

Start → Read A,B → SUM = A+B → Print SUM → Stop



*Flowchart for adding two numbers — shows the sequence of operations.*

**Malayalam support:** ുടും ുടും ുടും ുടും ുടും ുടും ുടും ുടും ുടും ുടും  
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## Practical: Components Identification & System Assembling

This practical session focuses on identifying computer components and assembling a desktop and laptop system.

### Key Components to Identify

- **Motherboard:** The main circuit board. Identify processor sockets, CPU support, power connectors, keyboard connectors, BIOS chip, bus slots, chipset, cache memory, co-processors, and CMOS battery.
- **Memory (RAM):** Identify DIMM/SO-DIMM modules, speed ratings, and capacity.
- **Hard Disk Drive (HDD) / SSD:** Identify connectors (SATA, IDE), power connectors, and storage capacity.
- **SMPS (Switched Mode Power Supply):** Power supply unit that converts AC to DC for the computer.
- **CMOS Battery:** Maintains system configuration and clock when the computer is off.
- **Other connectors:** Power connector, keyboard connector (PS/2 or USB), monitor connector (VGA, HDMI, DVI).

### Desktop Assembling Procedure

1. Prepare the case and install the power supply (SMPS).
2. Install the motherboard into the case, aligning with standoffs.
3. Install the CPU into the socket (align the triangle marker).
4. Attach the CPU cooler fan.
5. Install RAM modules into the appropriate slots.
6. Connect power cables from SMPS to motherboard (24-pin), CPU (4/8-pin), and drives.
7. Install HDD/SSD and connect SATA/data cables.
8. Connect front panel connectors (power switch, reset, LEDs, USB, audio).
9. Connect monitor, keyboard, and mouse.
10. Power on and observe the POST screen (Power-On Self-Test) and OS loading.
11. Check system configuration using BIOS/UEFI or system information tools.

### Laptop Assembling & Disassembling

- Identify the parts of a laptop: keyboard, trackpad, display, battery, RAM, HDD/SSD, WiFi card, cooling fan, motherboard.
- Practice assembling and disassembling with proper ESD precautions.
- Note the differences from desktop: integrated components, smaller form factor, specialised connectors.

 **Safety note:** Always disconnect the power supply before opening the computer case. Use an anti-static wrist strap to prevent electrostatic discharge (ESD) damage to components. Handle components by their edges, not by the pins or circuits.

**Malayalam support:** കമ്പ്യൂട്ടറിലെ ഓരോ ഓർഗനും ശ്രദ്ധയോടെ കൈകാര്യം ചെയ്യേണ്ടതാണ്. കമ്പ്യൂട്ടറിലെ ഓരോ ഓർഗനും ശ്രദ്ധയോടെ കൈകാര്യം ചെയ്യേണ്ടതാണ്. കമ്പ്യൂട്ടറിലെ ഓരോ ഓർഗനും ശ്രദ്ധയോടെ കൈകാര്യം ചെയ്യേണ്ടതാണ്. കമ്പ്യൂട്ടറിലെ ഓരോ ഓർഗനും ശ്രദ്ധയോടെ കൈകാര്യം ചെയ്യേണ്ടതാണ്. CPU, RAM, HDD, SMPS, CMOS എന്നിവയെ ശ്രദ്ധയോടെ കൈകാര്യം ചെയ്യേണ്ടതാണ്. POST എന്നിവയെ ശ്രദ്ധയോടെ കൈകാര്യം ചെയ്യേണ്ടതാണ്. POST എന്നിവയെ ശ്രദ്ധയോടെ കൈകാര്യം ചെയ്യേണ്ടതാണ്. POST എന്നിവയെ ശ്രദ്ധയോടെ കൈകാര്യം ചെയ്യേണ്ടതാണ്.

 **Module I Summary:** You have learned about computer characteristics, generations, components, memory hierarchy, hardware/software relationship, programming languages, translators, algorithms, flowcharts, and practical component identification and assembling. These form the foundation for all subsequent modules.

## MODULE II

### Operating Systems — Types, Functions & Installation

Understanding the core software that manages computer hardware and provides a platform for applications — from definitions to installation.

Theory: 11 hours

Practical: 8 hours

CO: CO2

Bloom: Apply



## Definition, Functions & Layers of Operating System

**Operating System (OS):** A system software that acts as an intermediary between the computer hardware and the user. It manages hardware resources, provides a user interface, and enables application software to run.

### Functions of an Operating System

- **Process Management:** Manages the execution of processes (programs). Includes scheduling, creation, and termination of processes.
- **Memory Management:** Allocates and manages memory for programs. Ensures efficient use of RAM.
- **File System Management:** Organises files and directories on storage devices. Manages read/write operations, permissions, and file organisation.
- **Device Management:** Controls I/O devices through device drivers. Handles communication between devices and the CPU.
- **Security & Protection:** Protects user data from unauthorised access. Manages user authentication and permissions.
- **User Interface:** Provides a way for users to interact with the computer (CLI or GUI).
- **Resource Allocation:** Distributes system resources (CPU time, memory, I/O) among processes.

### Layers of an Operating System

| Layer                       | Description                                                                              |
|-----------------------------|------------------------------------------------------------------------------------------|
| Hardware                    | Physical components: CPU, memory, I/O devices                                            |
| Kernel                      | Core of the OS, manages hardware, processes, memory, and files. Runs in privileged mode. |
| System Services             | Utilities, device drivers, file system management, networking stack.                     |
| Shell / Command Interpreter | Interface between the user and the OS (CLI or GUI).                                      |
| User Applications           | Application software running on top of the OS (MS Office, browsers, games).              |

OS Layers — from hardware to user

**Malayalam support:** ഓപ്പറേറ്റിംഗ് സിസ്റ്റം (OS) കമ്പ്യൂട്ടർ ഹാർഡ്വെയറും ഉപയോക്താവും തമ്മിൽ ഇടവേള നിറയ്ക്കുന്ന സോഫ്റ്റ്‌വെയറാണ്. അതിന്റെ പ്രധാന പ്രവർത്തനങ്ങൾ: പ്രോസസ്സ് മാനേജ്മെന്റ്, മെമറി മാനേജ്മെന്റ്, ഫയൽ സിസ്റ്റം മാനേജ്മെന്റ്, ഡിവൈസ് മാനേജ്മെന്റ്, സെക്യൂരിറ്റി പ്രൊട്ടക്ഷൻ, യൂസർ ഇന്റർഫേസ്. OS-ന്റെ ഘട്ടങ്ങൾ: ഹാർഡ്വെയർ → കേർണൽ → സിസ്റ്റം സർവീസ് → ഷെൽ → യൂസർ അപ്ലിക്കേഷൻ.

# Types of Operating Systems

Operating systems can be classified based on the number of users they support, the processing technique they use, and their response characteristics.

## Based on Users

- **Single-User OS:** Designed for one user at a time. Examples: MS-DOS, Windows 95/98.
- **Multi-User OS:** Allows multiple users to use the system simultaneously. Examples: Linux, UNIX, Windows Server.

## Based on Processing Technique

- **Batch Processing OS:** Jobs are collected in a batch and processed sequentially. No user interaction during processing. Example: Early mainframes.
- **Time-Sharing OS (Multitasking):** Multiple users share CPU time in small time slices. Gives the illusion of simultaneous execution. Examples: UNIX, Linux, Windows.
- **Real-Time OS (RTOS):** Responds to events in a guaranteed time frame. Used in embedded systems, robotics, and industrial control. Examples: VxWorks, QNX.

## Operating System Techniques

- **Multiprogramming:** Multiple programs are loaded into memory, and the CPU switches between them to keep the CPU busy.
- **Multitasking:** Similar to multiprogramming, but includes user interaction. Multiple tasks (processes) appear to run simultaneously.
- **Multiprocessing:** Uses multiple CPUs to execute multiple processes simultaneously. Requires special hardware.

| Type             | User Interaction | Response Time             | Example              |
|------------------|------------------|---------------------------|----------------------|
| Batch Processing | None (offline)   | High (hours/days)         | Early IBM mainframes |
| Time-Sharing     | Interactive      | Low (milliseconds)        | UNIX, Linux, Windows |
| Real-Time        | Event-driven     | Guaranteed (microseconds) | VxWorks, QNX         |

Comparison of OS Types

**Malayalam support:** OS-കൾ ഒന്നിനൊന്നിനായി വ്യത്യസ്തമായി തിരഞ്ഞെടുക്കാവുന്നതാണ് (single-user, multi-user), പ്രവർത്തന രീതിയെ അനുസരിച്ച് (batch, time-sharing, real-time), കമ്പ്യൂട്ടറുകളുടെ എണ്ണത്തെ അനുസരിച്ച് (multiprogramming, multitasking, multiprocessing) തിരഞ്ഞെടുക്കാവുന്നതാണ്.

## Popular Operating Systems — Linux, Windows & Mobile OS

### Microsoft Windows

- **History:** Started with Windows 1.0 (1985). Windows 95 introduced the Start menu. Windows 10/11 are the latest versions.
- **Features:** GUI-based, wide application support, user-friendly, extensive hardware compatibility, regular updates.
- **Editions:** Home, Pro, Enterprise, Education, Server.
- **File System:** NTFS (New Technology File System) — supports large files, security, and compression.

### Linux

- **History:** Created by Linus Torvalds in 1991 as a free and open-source alternative to UNIX.
- **Features:** Open-source, free, highly customisable, stable, secure, widely used in servers and embedded systems.
- **Distributions (Distros):** Ubuntu (user-friendly), Fedora, Debian, CentOS, Red Hat Enterprise Linux (RHEL).
- **File System:** ext4, XFS, Btrfs — supports advanced features like journaling and snapshots.

### Mobile Operating Systems (MOS)

- **Android:** Based on Linux kernel, open-source, developed by Google. Most widely used mobile OS. Features: customisable, large app ecosystem, supports multiple hardware vendors.
- **iOS:** Developed by Apple, closed-source, runs on iPhone and iPad. Features: highly secure, seamless ecosystem, curated app store, regular updates.
- **Others:** HarmonyOS (Huawei), KaiOS (feature phones), watchOS (Apple Watch).

### Key Features of Mobile OS

- Touch-based UI with gesture controls.
- App stores with millions of applications.
- Cloud integration for backup and sync.
- Location services (GPS), sensors (accelerometer, gyroscope).
- Security: biometric authentication (fingerprint, face ID), app permissions.

### Future Trends of Mobile OS

- **AI Integration:** On-device AI processing, intelligent assistants.
- **Foldable & Dual-Screen Support:** Optimised for new form factors.
- **Privacy & Security:** Increased focus on user data protection.
- **Cross-Device Integration:** Seamless experience across phone, tablet, laptop, and IoT devices.

**Malayalam support:** ഓപ്പറേറ്റിംഗ് OS-കൾ: Windows (GUI, wide app support), Linux (open-source, stable, server), Android (open-source, mobile), iOS (closed-source, secure). ഓപ്പറേറ്റിംഗ് OS-കൾക്ക്: AI, സുരക്ഷിതത്വം, സുരക്ഷിതത്വം, സുരക്ഷിതത്വം-സുരക്ഷിതത്വം സുരക്ഷിതത്വം.



## Practical: Installation of Windows, Linux & Application Software

This practical session covers the installation of Windows and Linux operating systems, as well as various application software packages.

### Installing Windows Operating System

1. Insert the Windows installation media (USB drive or DVD).
2. Boot from the installation media (enter BIOS/UEFI to change boot order).
3. Select language, time, and keyboard preferences.
4. Click "Install Now" and enter the product key (if required).
5. Select the operating system edition (e.g., Windows 11 Home/Pro).
6. Accept the license terms.
7. Choose the installation type: "Custom: Install Windows only (advanced)".
8. Select the drive/partition where Windows will be installed.
9. Follow the on-screen prompts for user account creation, privacy settings, and updates.
10. After installation, install device drivers and update the OS.

### Installing Linux Operating System (Ubuntu example)

1. Download the Ubuntu ISO file from the official website.
2. Create a bootable USB drive using tools like Rufus or BalenaEtcher.
3. Boot from the USB drive.
4. Select "Try Ubuntu" or "Install Ubuntu".
5. Choose language and keyboard layout.
6. Select "Normal installation" and "Download updates while installing" (optional).
7. Choose "Erase disk and install Ubuntu" or "Something else" for manual partitioning.
8. Select your timezone and create a user account.
9. Wait for the installation to complete and restart the system.
10. Remove the installation media and boot into the new Ubuntu system.

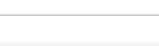
### Installing Application Software

- **MS Office:** Run the installer, accept the license, choose the installation type (typical/custom), and complete the setup.
- **Printer Software:** Connect the printer, run the driver installer, or use Windows/Linux built-in drivers.
- **Other Software:** Adobe Reader, web browsers, antivirus, media players, development tools.



**Common issues:** Ensure you have a valid product key for Windows. For Linux, check hardware compatibility (especially WiFi and graphics drivers). Always back up important data before installing a new OS.

 **Safety note:** Keep your system plugged in during OS installation to avoid power interruption. Use genuine installation media to avoid malware. After installation, install security updates and antivirus software.

**Malayalam support:** Windows, Linux OS-   .   , BIOS/UEFI- boot order    . MS Office, printer drivers     .

 **Module II Summary:** You have learned about the definition, functions, and layers of an operating system, different types of OS (single/multi-user, batch, time-sharing, real-time), OS techniques (multiprogramming, multitasking, multiprocessing), popular OS (Windows, Linux, Android, iOS), and practical installation procedures.

### MODULE III

## Data Processing & Communication

Understanding how data is processed, stored, and transmitted — from file management to communication media and modems.

Theory: 11 hours

Practical: 8 hours

CO: CO3

Bloom: Apply

## Data Processing — Definition, Types & File Management

**Data Processing:** The process of collecting, manipulating, and converting raw data into meaningful information. It involves input, processing, and output stages.

### Types of Data Processing

- **Batch Processing:** Data is collected over time and processed in batches. Used for payroll, billing, and reporting.
- **Real-Time Processing:** Data is processed immediately as it is received. Used for ATM transactions, online booking, and monitoring systems.
- **Online Processing:** Interactive processing where the user waits for a response. Used for data entry and query systems.
- **Distributed Processing:** Data is processed across multiple computers. Used in cloud computing and large-scale systems.

### File Management System

- **File:** A collection of related data stored on a storage device.
- **File Types:**
  - **Master File:** Contains permanent data (e.g., employee records, customer details).
  - **Transaction File:** Contains temporary data for processing (e.g., daily sales, daily attendance).
  - **Reference File:** Contains lookup data (e.g., price list, code tables).
  - **Report File:** Contains formatted output for printing or viewing.
- **File Operations:** Create, read, write, update, delete, rename, copy, move, backup.

### Database Management System (DBMS)

- **Definition:** A software system that manages the storage, retrieval, and manipulation of structured data.
- **Features:** Data security, integrity, concurrency control, backup and recovery, query language (SQL).
- **Applications:** Banking systems, hospital management, railway reservations, e-commerce, social media.
- **Difference from File System:** DBMS reduces redundancy, ensures data consistency, provides concurrent access, and enforces security rules.

| Feature           | File System                    | DBMS                         |
|-------------------|--------------------------------|------------------------------|
| Data Redundancy   | High (duplicate data possible) | Low (normalised)             |
| Data Consistency  | Difficult to maintain          | Automatically maintained     |
| Concurrent Access | Limited or manual locking      | Built-in concurrency control |

| Feature        | File System              | DBMS                            |
|----------------|--------------------------|---------------------------------|
| Security       | Basic (file permissions) | Granular (user/role-based)      |
| Query Language | No standard              | SQL (Structured Query Language) |

File System vs DBMS

**Malayalam support:** കേവലം ക്രുഡ് ഡാറ്റ: raw data-എന്ന അർത്ഥത്തിൽ വിവരങ്ങൾ. ക്രുഡ്: batch, real-time, online, distributed. തരം തിരിവ്: master, transaction, reference, report. DBMS ഉപയോഗിക്കുന്ന സ്ഥാപനങ്ങൾ (bank, hospital, railway).





## Practical: Performing Data Processing (Excel Example)

This practical session involves collecting raw data, processing it in Microsoft Excel, and generating formatted reports.

### Example: Daily Sales of a Shop for 5 Days

1. **Input:** Enter the daily sales data for 5 days (e.g., product name, quantity sold, price per unit).
2. **Processing:**
  - Calculate total sales for each day (quantity × price).
  - Calculate the average sales over the 5 days.
  - Sort the data by product name or by total sales.
3. **Output:** Generate a formatted report with tables, charts, and summary statistics.

#### ■ Sample Calculation

**Day 1:** Product A — 10 units × ₹50 = ₹500

**Day 2:** Product A — 12 units × ₹50 = ₹600

**Day 3:** Product A — 8 units × ₹50 = ₹400

**Total:** ₹500 + ₹600 + ₹400 = ₹1,500

**Average:** ₹1,500 ÷ 3 = ₹500 per day

### File Operations

- **Create:** New file (e.g., sales\_data.xlsx).
- **Rename:** Change file name (e.g., sales\_report\_2026.xlsx).
- **Copy:** Make a duplicate for backup.
- **Move:** Transfer to a different folder.
- **Delete:** Remove unwanted files.
- **Backup:** Copy to pen drive or cloud storage (Google Drive, OneDrive).

**Malayalam support:** Excel ക്രമീകരണ പ്രവർത്തനങ്ങൾ: പുതിയ ഫയൽ സൃഷ്ടിക്കുക, ഫയൽ പേര് മാറ്റുക (total, average, sort) കണക്കാക്കുക, കോപ്പി ചെയ്യുക, മൂവ് ചെയ്യുക, ഡിളീറ്റ് ചെയ്യുക. ഈ പ്രവർത്തനങ്ങൾ: create, rename, copy, move, delete, backup.

## Data Communication — Elements, Modes & Signals

**Data Communication:** The exchange of data between two devices via a transmission medium. It is fundamental to networking and the internet.

### Elements of a Communication System

- **Sender (Source):** The device that sends the data (e.g., computer, smartphone).
- **Receiver (Destination):** The device that receives the data.
- **Message:** The data being transmitted (text, audio, video, etc.).
- **Transmission Medium:** The physical path through which data travels (wired: copper, fibre; wireless: radio, microwave).
- **Protocol:** A set of rules that governs the communication (e.g., TCP/IP, HTTP).

### Modes of Data Communication

- **Simplex:** One-way communication. The sender can only send, and the receiver can only receive. Example: Radio broadcast, TV transmission.
- **Half-Duplex:** Two-way communication, but only one device can send at a time. Example: Walkie-talkie, CB radio.
- **Full-Duplex:** Two-way communication where both devices can send and receive simultaneously. Example: Telephone, video conferencing.

### Analog and Digital Signals

- **Analog Signal:** Continuous wave signal that varies in amplitude and frequency. Used in traditional telephone lines, radio, and TV. Susceptible to noise and attenuation.
- **Digital Signal:** Discrete signal with two states (0 and 1). Used in modern computer networks, fibre optics, and digital communication. More immune to noise and easier to process.

| Feature              | Analog                         | Digital                         |
|----------------------|--------------------------------|---------------------------------|
| Signal Type          | Continuous wave                | Discrete (0/1)                  |
| Noise Susceptibility | High                           | Low                             |
| Processing           | Complex                        | Simple (computers)              |
| Example              | Radio, telephone (traditional) | Computer networks, fibre optics |

*Comparison: Analog vs Digital Signals*

**Malayalam support:** കമ്മ്യൂണിക്കേഷൻ സിസ്റ്റം — സെൻഡർ, റീസീവർ, മെസ്സേജ്, മീഡിയം, പ്രോട്ടോക്കോൾ.  
കമ്മ്യൂണിക്കേഷൻ: സെൻഡർ, റീസീവർ, മെസ്സേജ്, മീഡിയം, പ്രോട്ടോക്കോൾ.  
കമ്മ്യൂണിക്കേഷൻ: simplex (one-way), half-duplex (alternate), full-duplex (simultaneous). Signals: analog (continuous) vs digital (discrete 0/1).

## Transmission Media — Guided & Unguided

**Transmission Media** is the physical or wireless path that carries data from sender to receiver. It is broadly classified into guided (wired) and unguided (wireless).

### Guided Media (Wired)

- **Twisted Pair Cable:**
  - Two insulated copper wires twisted together to reduce electromagnetic interference.
  - Used in telephone networks, Ethernet (10/100/1000BASE-T).
  - Types: Shielded (STP) and Unshielded (UTP).
  - Advantages: Low cost, flexible, easy to install.
  - Disadvantages: Limited distance, susceptible to interference.
- **Coaxial Cable:**
  - Has a central conductor surrounded by insulation, a braided shield, and an outer jacket.
  - Used in cable TV, broadband internet, and CCTV systems.
  - Advantages: Better shielding, longer distance than twisted pair.
  - Disadvantages: Thicker, more expensive, harder to install.
- **Optical Fibre Cable:**
  - Uses light pulses to transmit data through a glass or plastic core.
  - Used in high-speed internet, backbone networks, and telecommunications.
  - Advantages: Very high bandwidth, immune to electromagnetic interference, long distance (up to 100 km).
  - Disadvantages: High cost, fragile, complex installation.

### Unguided Media (Wireless)

- **Radio Waves:** Omni-directional, used in FM/AM radio, Bluetooth, WiFi (2.4 GHz, 5 GHz).
- **Microwaves:** Directional, used in satellite communication, point-to-point links, and radar.
- **Infrared:** Short-range, line-of-sight, used in remote controls and IR communication.
- **Li-Fi (Light Fidelity):** Uses visible light for communication, very high speed.

| Media            | Bandwidth        | Distance        | Noise Immunity | Cost   |
|------------------|------------------|-----------------|----------------|--------|
| Twisted Pair     | Up to 1 Gbps     | ~100 m          | Low            | Low    |
| Coaxial          | Up to 1 Gbps     | ~500 m          | Medium         | Medium |
| Optical Fibre    | 100+ Gbps        | ~100 km         | High           | High   |
| Wireless (Radio) | ~1 Gbps (WiFi 6) | ~100 m (indoor) | Low            | Medium |

**Malayalam support:** നയിക്കുന്ന മാധ്യമം — guided (wired: twisted pair, coaxial, optical fibre) നയിക്കുന്ന (wireless: radio, microwave, infrared) നയിക്കുന്ന. നയിക്കുന്ന മാധ്യമം നയിക്കുന്ന, bandwidth, distance, cost നയിക്കുന്ന നയിക്കുന്ന.

## Modems & Wireless Data Connectivity

**Modem (Modulator-Demodulator):** A device that converts digital signals from a computer into analog signals for transmission over telephone lines (modulation) and converts incoming analog signals back to digital (demodulation).

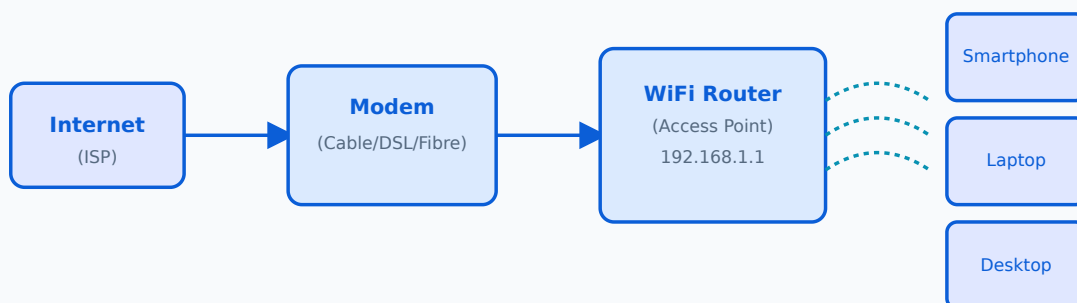
### Types of Modems

- **Dial-up Modem:** Uses standard telephone lines (PSTN). Slow speeds (up to 56 kbps). Now mostly obsolete.
- **ADSL Modem:** Uses telephone lines but with higher speeds (up to 24 Mbps). Uses frequency division to separate voice and data.
- **Cable Modem:** Uses coaxial cable TV networks for internet access. Speeds up to 1 Gbps.
- **Fibre Optic Modem (ONT):** Used with fibre-to-the-home (FTTH) connections. Very high speeds (up to 1 Gbps+).
- **Wireless Modem (4G/5G):** Uses cellular networks for internet access. Integrated into smartphones or external dongles.

### Basics of Wireless Data Connectivity

- **WiFi (Wireless Fidelity):** Based on IEEE 802.11 standards. Uses radio waves (2.4 GHz and 5 GHz bands).
- **Bluetooth:** Short-range communication (up to 10 m). Used for connecting peripherals (keyboards, headphones).
- **Cellular Networks:** 4G (LTE), 5G — provide mobile internet with wide coverage.
- **Components:** Wireless Access Point (WAP), Router, Network Interface Card (NIC), Antenna.

Internet → Modem → WiFi Router → Smartphone / Laptop / Desktop



*A typical WiFi network: Internet → Modem → WiFi Router → Devices (smartphone, laptop, desktop).*

**Malayalam support:** ഡിജിറ്റൽ — digital to analog (modulation), analog to digital (demodulation) ഡിജിറ്റൽ കമ്മ്യൂണിക്കേഷൻ. ഡിജിറ്റൽ: dial-up, ADSL, cable, fibre, wireless (4G/5G). WiFi (wireless fidelity) radio waves ഉപയോഗിക്കുന്ന ഉപകരണങ്ങൾ-റൂട്ടർ, അക്സസ് പോയിന്റ്, NIC ഉപയോഗിക്കുന്ന ഉപകരണങ്ങൾ.



## Practical: Guided Media Demonstration & UTP Cable Crimping

This practical session involves identifying guided media (cables), understanding modems and WiFi components, and performing UTP cable crimping.

### Demonstration of Guided Media

- **Twisted Pair Cable (UTP/STP):** Observe the colour-coded wires, the RJ45 connector, and the cable jacket. Understand the T568A and T568B wiring standards.
- **Coaxial Cable:** Observe the central conductor, insulation, braided shield, and outer jacket. Identify the BNC or F-type connector.
- **Optical Fibre Cable:** Observe the glass core, cladding, buffer coating, and connector (SC, LC, ST). Note that it uses light pulses, not electrical signals.
- **Modem & WiFi Components:** Identify different types of modems (ADSL, cable, fibre), wireless access points, and antennas.

### UTP Cable Crimping

**Tools required:** UTP cable, RJ45 connectors, crimping tool, cable stripper, cable tester.

1. Strip about 2.5 cm of the outer jacket from the cable end using the cable stripper.
2. Untwist the four pairs of wires and arrange them according to the T568B standard (or T568A).
3. **T568B wiring order (from left to right, with clip facing down):**  
Orange/White, Orange, Green/White, Blue, Blue/White, Green, Brown/White, Brown.
4. Cut the wires straight across, leaving about 1.25 cm from the jacket.
5. Insert the wires into the RJ45 connector (ensure the wires go all the way to the end).
6. Place the connector into the crimping tool and crimp firmly.
7. Repeat the process on the other end of the cable.
8. Test the cable using a cable tester to ensure continuity and correct pin configuration.

 **Common mistakes:** Incorrect wire order, wires not reaching the end of the connector, stripping too much jacket, crimping too lightly. Always double-check the T568B/T568A standard before crimping.

 **Safety note:** Use proper tools and handle the cable cutter carefully. Do not pull cables with excessive force. Test cables before using them in a live network.



**Malayalam support:** UTP കേബിൾ കോഡിംഗ് സ്കീം: T568B കോഡിംഗ്

കോഡിംഗ് സ്കീം കേബിൾ RJ45 കോൺക്ടർ കോഡിംഗ് സ്കീം. കോഡിംഗ് സ്കീം  
കാൽ, കേബിൾ കോഡിംഗ് സ്കീം, കേബിൾ കോഡിംഗ് സ്കീം കോഡിംഗ് സ്കീം. കോഡിംഗ്  
സ്കീം: Orange/White, Orange, Green/White, Blue, Blue/White, Green,  
Brown/White, Brown.



**Module III Summary:** You have learned about data processing types, file management, DBMS, elements of data communication, communication modes (simplex, half-duplex, full-duplex), analog vs digital signals, transmission media (guided and unguided), modems, wireless connectivity, and practical cable crimping.

#### MODULE IV

### Computer Networks & Topologies

Understanding how computers communicate, network topologies, components, internet basics, and cloud storage.

Theory: 11 hours

Practical: 8 hours

CO: CO4

Bloom: Apply

## Computer Communication Networks — Definition & Advantages

**Computer Communication Network:** A set of interconnected computers and devices that communicate with each other using a common set of protocols. The goal is to share resources, data, and services.

### Advantages of Networks

- **Resource Sharing:** Share hardware (printers, scanners), software, and data across the network.
- **Communication:** Email, instant messaging, video conferencing, VoIP.
- **Centralised Data Management:** Data stored on servers can be accessed from any connected device.
- **Cost Efficiency:** Reduces hardware costs by sharing devices and storage.
- **Reliability:** If one device fails, the network can still function (redundancy).
- **Scalability:** Networks can be expanded to include new devices and users.
- **Remote Access:** Users can access network resources from anywhere (VPN, remote desktop).
- **Security:** Centralised security management (firewalls, authentication, encryption).

**Malayalam support:** കമ്പ്യൂട്ടർ കമ്മ്യൂണിക്കേഷൻ നെറ്റ്വർക്ക് — വിവിധ കമ്പ്യൂട്ടറുകൾ  
ഒരുമിച്ച് വിവിധ വിഭവങ്ങൾ (പ്രിന്ററുകൾ, ഡാറ്റ) പങ്കിടുന്നതിനും ആശയവിനിമയത്തിനും (ഇമെയിൽ, വീഡിയോ),  
കേന്ദ്രീകൃതമായ മാനേജ്മെന്റും, ചെലവ് കുറയ്ക്കലും, വിശ്വസ്തതയും, സ്കെയിലബിലിറ്റിയും, റിമോട്ട് അക്സസ്സും,  
സുരക്ഷയും ഉറപ്പുവരുത്തുന്നു. **Malayalam support:** resource sharing  
(printers, data), communication (email, video), centralised management,  
cost efficiency, reliability, scalability, remote access, security.

## Types of Networks — PAN, LAN, MAN, WAN

Networks are classified based on their size, geographic coverage, and the technology used.

- **PAN (Personal Area Network):**
  - Covers a very small area (a few meters).
  - Used for connecting personal devices: smartphone, laptop, headphones, smartwatch.
  - Technologies: Bluetooth, USB, Infrared.
  - Example: Connecting a Bluetooth headset to a phone.
- **LAN (Local Area Network):**
  - Covers a limited area: a room, a building, a campus.
  - Typically uses Ethernet (wired) and WiFi (wireless).
  - High speed, low latency, private ownership.
  - Example: A school computer lab, office network, home WiFi.
- **MAN (Metropolitan Area Network):**
  - Covers a city or a metropolitan area.
  - Combines multiple LANs in a city.
  - Examples: Cable TV networks, city-wide WiFi, municipal networks.
- **WAN (Wide Area Network):**
  - Covers a large geographic area (country, continent, global).
  - Connects multiple LANs and MANs.
  - Examples: The Internet, corporate networks connecting branch offices.
  - Technologies: Leased lines, satellite, fibre optics, cellular networks.

| Type | Size            | Speed            | Example             | Technology       |
|------|-----------------|------------------|---------------------|------------------|
| PAN  | Few meters      | Low-Medium       | Bluetooth headset   | Bluetooth, USB   |
| LAN  | Building/Campus | High (1-10 Gbps) | School computer lab | Ethernet, WiFi   |
| MAN  | City            | Medium-High      | Cable TV network    | Fibre, microwave |
| WAN  | Country/Global  | Varies           | The Internet        | Satellite, fibre |

Comparison: PAN, LAN, MAN, WAN

**Malayalam support:** പേര്: PAN (personal — Bluetooth), LAN (local — office/school), MAN (metropolitan — city), WAN (wide — internet). ഉപയോഗം, വേഗം, ഉദാഹരണം എന്നിവയിലാണ് വ്യത്യാസം.

## Network Topologies — Bus, Ring, Mesh, Star

**Network Topology:** The arrangement or physical layout of devices (nodes) and connections in a network. Each topology has its own advantages and disadvantages.

### Bus Topology

- All devices are connected to a single central cable (bus).
- Data travels in both directions along the bus.
- Advantages: Simple, cost-effective, easy to extend.
- Disadvantages: Single point of failure (cable break), limited distance, performance degrades with more devices.

### Ring Topology

- Each device is connected to two others, forming a ring.
- Data travels in one direction around the ring.
- Advantages: Predictable performance, no collisions.
- Disadvantages: Single break breaks the entire ring, difficult to troubleshoot.

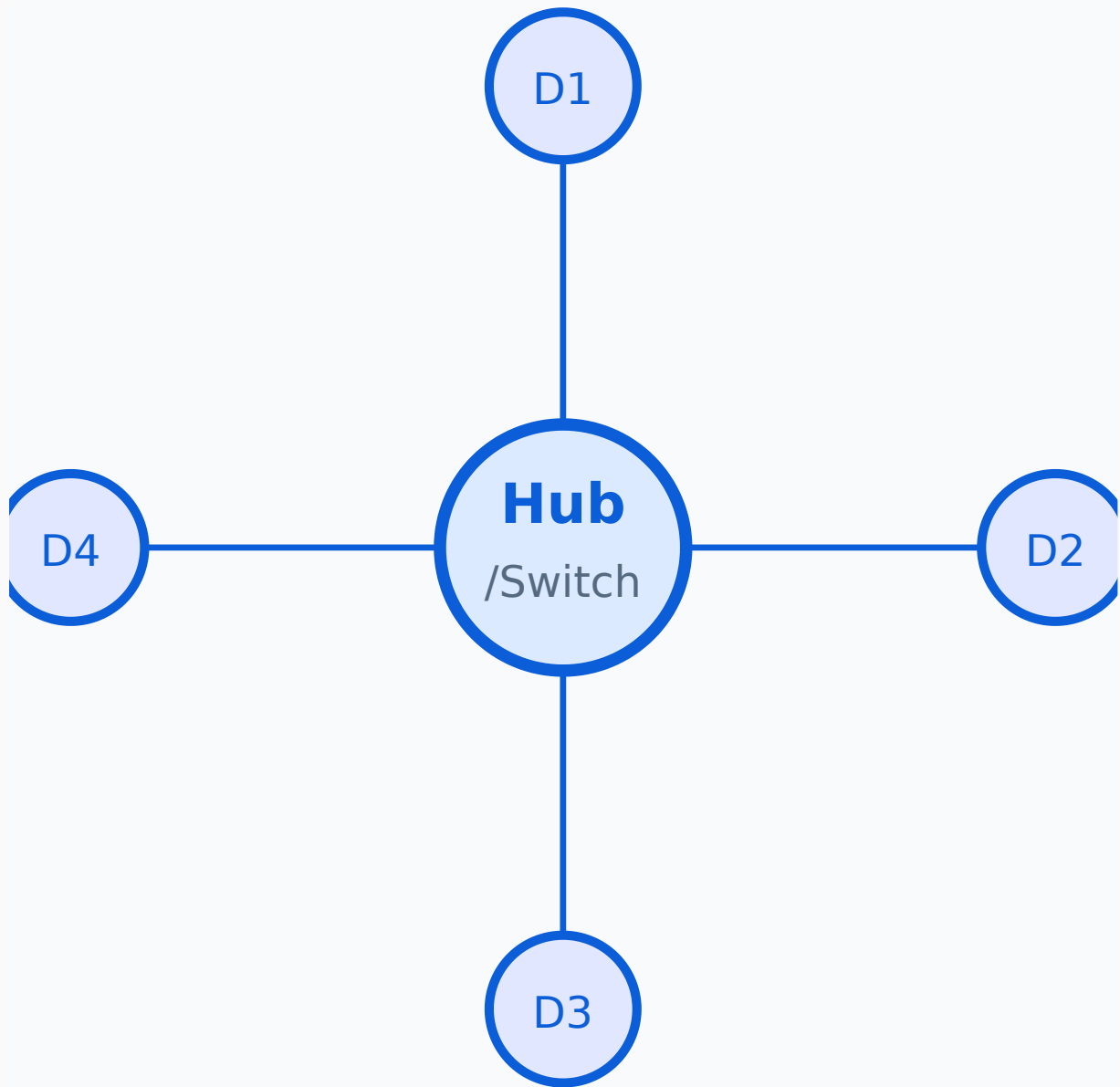
### Mesh Topology

- Every device is connected to every other device (full mesh) or partially connected (partial mesh).
- Advantages: High redundancy, multiple paths, robust.
- Disadvantages: Expensive, complex cabling, difficult to manage.

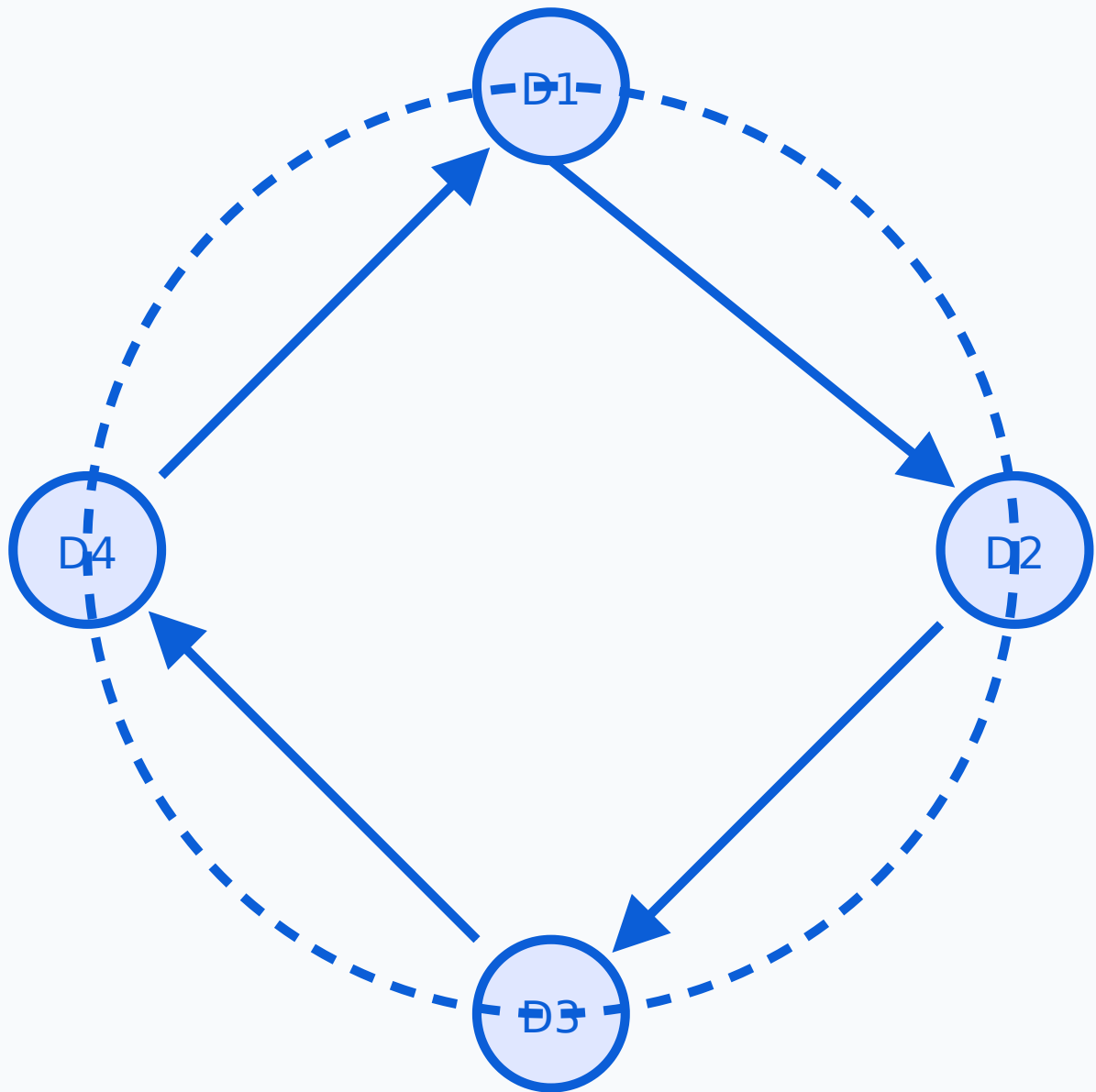
### Star Topology

- All devices are connected to a central hub or switch.
- Advantages: Easy to manage, centralised control, failure of one device does not affect others.
- Disadvantages: Single point of failure (central hub/switch), high cabling cost.

Central hub/switch connected to multiple devices



Devices connected in a ring, data flows in one direction



| Topology    | Advantages                  | Disadvantages                                | Applications                         |
|-------------|-----------------------------|----------------------------------------------|--------------------------------------|
| <b>Bus</b>  | Simple, low cost            | Single point of failure, limited scalability | Small networks, early Ethernet       |
| <b>Ring</b> | Predictable, no collisions  | Single break breaks the network              | Token Ring (legacy), SONET           |
| <b>Star</b> | Easy to manage, centralised | Hub/switch is a single point of failure      | Most modern LANs (Ethernet)          |
| <b>Mesh</b> | Highly redundant, robust    | Expensive, complex cabling                   | Critical systems, military, backbone |

Comparison of Network Topologies

**Malayalam support:** വാട്സ്ആപ്പ് വാട്സ്ആപ്പ്: Bus (single cable), Ring (circle), Star (central hub), Mesh (fully connected). വാട്സ്ആപ്പ് വാട്സ്ആപ്പ്. Star വാട്സ്ആപ്പ് വാട്സ്ആപ്പ് (Ethernet LAN-വാട്സ്ആപ്പ്).

Network Components — Hub, Switch, Router, Gateway

- **Hub:** A simple network device that broadcasts data to all connected devices. Works at Layer 1 (Physical layer) of the OSI model. Inefficient, but inexpensive. Mostly obsolete in modern networks.
- **Switch:** A smarter network device that directs data only to the intended recipient. Works at Layer 2 (Data Link layer). Uses MAC addresses to forward frames. Efficient, widely used.
- **Router:** Connects different networks (e.g., LAN to WAN). Works at Layer 3 (Network layer). Uses IP addresses to route packets. Determines the best path for data.
- **Gateway:** A device that connects networks with different protocols. Translates between different architectures. Works at multiple OSI layers. Example: A router that connects a LAN to the internet is also a gateway.
- **Other Components:** Repeater (amplifies signals), Bridge (connects two LANs), Modem (modulates/demodulates).

| Device | Layer     | Operation                | Intelligence | Example Use           |
|--------|-----------|--------------------------|--------------|-----------------------|
| Hub    | Physical  | Broadcasts to all        | None (dumb)  | Legacy networks       |
| Switch | Data Link | Forwards to specific MAC | Smart        | Modern LANs           |
| Router | Network   | Routes between networks  | Very smart   | Internet connectivity |

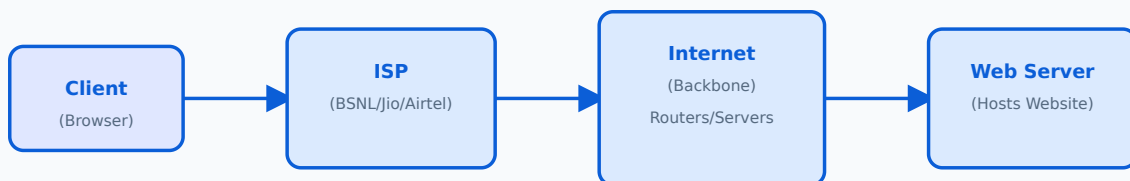
Comparison: Hub, Switch, Router

**Malayalam support:** വാട്സ്ആപ്പ് വാട്സ്ആപ്പ്: Hub (broadcast to all), Switch (directs to specific device using MAC), Router (connects different networks using IP), Gateway (protocol translation). Hub-വാട്സ്ആപ്പ് Switch-വാട്സ്ആപ്പ് Router-വാട്സ്ആപ്പ് intelligent വാട്സ്ആപ്പ്.

## Internet Basics — ISP, Web Server, Browser, IP Address, Domain Names, Intranet

- **ISP (Internet Service Provider):** A company that provides internet access to customers. Examples: BSNL, Jio, Airtel, Verizon.
- **Web Server:** A computer that hosts websites and serves web pages to clients. Responds to HTTP/HTTPS requests.
- **Web Browser:** A software application used to access and view web pages. Examples: Chrome, Firefox, Edge, Safari.
- **IP Address (Internet Protocol Address):** A unique numerical label assigned to each device on a network. IPv4: 192.168.1.1, IPv6: 2001:0db8:85a3::8a2e:0370:7334.
- **Domain Name:** A human-readable name for an IP address (e.g., google.com, polytech.edu). Resolved by DNS (Domain Name System).
- **Intranet:** A private network within an organisation that uses internet technologies but is restricted to authorised users. Example: A company's internal portal.

Client → ISP → DNS → Web Server → Client



**Malayalam support:** ഇന്റർനെറ്റ് സേവകൻ: ISP (service provider), Web Server (website host), Browser (client), IP Address (unique ID), Domain Name (human-readable), Intranet (private network). DNS domain names-IP addresses-ഇന്റർനെറ്റ് സേവകൻ.



## World Wide Web, Search Engines, Chatbots & Online Stores

- **World Wide Web (WWW):** A system of interlinked documents and resources accessed via the internet. Uses HTTP/HTTPS protocol. Invented by Tim Berners-Lee in 1989.
- **Search Engines:** Tools that search the web and return relevant results. Examples: Google, Bing, Yahoo, DuckDuckGo.
- **Search Engine Tips:**
  - **Keywords:** Use specific words. Example: "diploma computer applications syllabus".
  - **Quotation Marks (" "):** Search for exact phrases. Example: "Kerala Polytechnic revision 2026".
  - **Site Search (site:):** Limit results to a specific website. Example: site:sitttr.kerala.gov.in "computer applications".
  - **Minus (-):** Exclude words. Example: "computer applications -management".
- **Chatbots:** AI-powered programs that simulate human conversation. Examples: ChatGPT, Gemini, Copilot. Used for customer service, education, and assistance.
- **Online Stores (E-commerce):** Websites where products and services are bought and sold. Examples: Amazon, Flipkart, eBay, GoDaddy (domain/hosting).
- **Amazon Cloud Storage:** Amazon Web Services (AWS) provides cloud storage solutions (S3, EC2). Enables file storage, backup, and web hosting in the cloud.

 **Note:** The syllabus mentions "Basics of Amazon Cloud Storage" and "GoDaddy". These are introduced as examples of cloud services and domain/hosting providers. Amazon S3 (Simple Storage Service) and GoDaddy (domain registration and web hosting) are industry examples used for educational illustration.

**Malayalam support:** WWW — HTTP/HTTPS ന്റെ അടിസ്ഥാനപരമായ സങ്കേതം. Search Engines (Google, Bing) keywords, quotation marks, site: search സാധനങ്ങൾ ഉപയോഗിക്കുന്നു. Chatbots (AI), Online Stores (e-commerce), Amazon Cloud Storage (S3, EC2) സാധനങ്ങൾ ഉപയോഗിക്കുന്നു.



## Practical: Network Troubleshooting, Components & Commands

This practical session focuses on basic network troubleshooting, demonstrating network components, printer connection, and using network commands.

### Basic Network Troubleshooting

1. **Check Physical Connections:** Ensure cables are properly connected, lights on the switch/router are on.
2. **Restart Devices:** Power cycle the modem, router, and computer.
3. **Check IP Address:** Use `ipconfig` (Windows) or `ifconfig` / `ip a` (Linux) to verify network configuration.
4. **Ping Command:** Use `ping` to test connectivity to another device.
  - `ping 192.168.1.1` — test connectivity to the router.
  - `ping google.com` — test internet connectivity.
5. **Check Network Adapter:** Ensure the network adapter is enabled.
6. **Check DNS Settings:** Use `nslookup` to test DNS resolution.

### Network Commands

- **ipconfig (Windows):** Displays IP address, subnet mask, default gateway.
- **ifconfig / ip a (Linux):** Similar to ipconfig, shows network interface information.
- **ping:** Tests connectivity to a host.
- **tracert (Windows) / traceroute (Linux):** Traces the path to a destination.
- **nslookup:** Queries DNS for domain name resolution.
- **netstat:** Shows network connections and routing tables.

### Demonstrating Network Components

- **Client Computers:** Workstations that request resources from the network.
- **Switch / Hub:** Interconnect multiple devices in a LAN.
- **Router:** Connects the LAN to the internet (or other networks).
- **Network Cables:** UTP, fibre optic, coaxial — connect devices physically.
- **Printer Connection:** Connect a network printer via IP address or shared printer. Demonstrate how to add a network printer and print a test page.
- **Internet Sharing:** Share an internet connection across multiple devices using a router or software-based ICS.

### **Example: Finding Your IP Address**

**Windows:** Open Command Prompt → type ipconfig → look for "IPv4 Address".

**Linux:** Open Terminal → type ip a → look for "inet" under the active interface (e.g., eth0, wlan0).

#### **Example output (Windows):**

```
IPv4 Address. . . . . : 192.168.1.100
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
```

 **Safety note:** Do not run network troubleshooting commands on networks you do not own or have permission to access. Use `ping` responsibly — it should only be used for diagnostic purposes.

**Malayalam support:** ശാരീരിക ബന്ധനങ്ങൾ: physical connections, restart devices, IP address check (ipconfig/ifconfig), ping, tracert, nslookup. Network components: client computers, switch/hub, router, network cables, printer connection. ശൃംഖല ബന്ധനം network connectivity ശാസ്ത്രങ്ങൾ search engines ശാസ്ത്രങ്ങൾ ശാസ്ത്രങ്ങൾ.

 **Module IV Summary:** You have learned about computer networks, advantages, types (PAN, LAN, MAN, WAN), topologies (bus, ring, mesh, star), components (hub, switch, router, gateway), internet basics (ISP, web server, browser, IP, domain, intranet), WWW, search engines, chatbots, online stores, cloud storage, and practical troubleshooting techniques.

## QUICK REFERENCE

### Formula Bank

Key formulas, units, and equations from all modules — useful for revision and problem-solving.

$$\text{Speed} = \text{Distance} / \text{Time}$$

Data transfer rate (bps)

$$\text{Total Sales} = \Sigma (\text{Qty} \times \text{Price})$$

Aggregate sales calculation

$$\text{Average} = \text{Total} / \text{Count}$$

Mean of data points

$$\text{Bandwidth} = \text{Data Size} / \text{Time}$$

Network capacity (bps)

**IP Address: 192.168.x.x**

Private IPv4 address range

$$\text{Ping} = \text{Request} / \text{Response}$$

Network round-trip time (ms)

#### QUICK VISUAL REFERENCE

## Diagram Library

Key diagrams from each module for rapid revision.

### **Module I**

Computer block diagram, memory hierarchy, flowchart symbols, flowchart for adding two numbers.

### **Module II**

OS layers diagram, installation flow, Windows/Linux boot screens (conceptual).

### **Module III**

WiFi network setup, analog vs digital signals, guided media (twisted pair, coaxial, fibre).

### **Module IV**

Star topology, ring topology, internet flow diagram, hub/switch/router comparison.

#### PRACTICE & EXPLORE

## Self-Learning Activities

Official self-learning activities from the syllabus — complete these to reinforce your understanding.

## Module I Activities

### Week 1 — Class summary + Mind map

Draw a mind map showing characteristics of computers with real-life examples (speed → ATM, accuracy → billing). Write record books.

 5.5 hours

### Week 2 — Compare memory types

Prepare a table comparing RAM, ROM, Cache, Secondary Memory (speed, volatility, use).

 5.5 hours

### Week 3 — Classify hardware & software

List all hardware devices at home and classify as input/output/storage. Classify software you use daily into system, application, utility.

 5.5 hours

### Week 4 — Compare OS & create flowcharts

Compare Windows vs Linux or MS Office vs LibreOffice. Write algorithms and draw flow charts for sample problems.

 5.5 hours

## Module II Activities

### Week 5 — OS functions & examples

Match OS functions with real examples (process management → multitasking). Prepare module notes.

 5.5 hours

### Week 6 — OS types chart

Prepare a chart showing batch, time-sharing, real-time OS. Write record books.

 5.5 hours

### **Week 7 — Compare mobile OS**

Compare Android and iOS (features, openness, apps).

 5.5 hours

### **Week 8 — Module summary**

Summarise Module II, write record books.

 5.5 hours

## **Module III Activities**

### **Week 9 — Input → Process → Output → Storage**

Draw the IPOS cycle with examples. Class summary presentation. Write record books.

 5.5 hours

### **Week 10 — File organisation**

Organise files on your computer into folders and note the benefits.

 5.5 hours

### **Week 11 — Database identification**

Identify where databases are used (bank, hospital, railway). Differentiate file system vs DBMS.

 5.5 hours

### **Week 12 — Communication modes & media**

Prepare a table comparing simplex, half-duplex, full-duplex with examples. Compare twisted pair, coaxial, fibre, radio, microwave. Illustrate unguided media.

 5.5 hours

## **Module IV Activities**

### Week 13 — Network types chart

Prepare a chart showing LAN, MAN, WAN with examples. Write record books.

 5.5 hours

### Week 14 — Identify network components

Identify router, switch, modem at home or institution. Prepare module notes.

 5.5 hours

### Week 15 — Internet services & module summary

List 5 educational uses of the internet and 5 services (email, WWW, cloud). Prepare module summary. Write record books.

 5.5 hours

 **Total self-learning hours:** 82.5 hours across 15 weeks, as per the official syllabus.

#### EXAM STRUCTURE

### Assessment Methodology

Official CIE and ESE assessment scheme, series test pattern, and evaluation criteria.

#### Continuous Internal Evaluation (CIE) — 40 Marks

| Mode             | Description                                          | Marks     |
|------------------|------------------------------------------------------|-----------|
| Attendance       | Class attendance and punctuality                     | 5         |
| CA1              | Average of self-learning activities from each module | 10        |
| CA2              | Continuous evaluation                                | 5         |
| CA3              | Practical test (first half of experiments)           | 10        |
| CA4              | Practical test (second half of experiments)          | 10        |
| CA5              | Series Exam (2 modules)                              | 10        |
| <b>Total CIE</b> |                                                      | <b>40</b> |

*CIE assessment components*

#### End Semester Examination (ESE) — 60 Marks

| Mode             | Description                   | Marks     |
|------------------|-------------------------------|-----------|
| Theory           | Written exam (theory portion) | 20        |
| Practical        | Lab examination (internal)    | 40        |
| <b>Total ESE</b> |                               | <b>60</b> |

*ESE components*

**Note:** CIE + ESE = 100 marks total. The practical component carries significant weight (40 marks), reflecting the practicum nature of the course.

### Series Examination Pattern (Theory)

| Part         | Type                                     | Marks             |
|--------------|------------------------------------------|-------------------|
| Part A       | Very short answer (1 mark each)          | $4 \times 1 = 4$  |
| Part B       | Short answer (3 marks each)              | $4 \times 3 = 12$ |
| Part C       | Long answer (7 marks each) — with choice | $2 \times 7 = 14$ |
| <b>Total</b> |                                          | <b>25</b>         |

*Series exam (2 hours) — 25 marks*

### ESE Question Paper Pattern (Theory — 2.5 hours)

| Part         | Type                                        | Questions                    | Marks             |
|--------------|---------------------------------------------|------------------------------|-------------------|
| Part A       | 1 mark each                                 | 2 from each module (total 8) | $8 \times 1 = 8$  |
| Part B       | 3 marks each (2 out of 3 per module)        | $2 \times 4 = 8$             | $8 \times 3 = 24$ |
| Part C       | 7 marks each (1 set with choice per module) | $1 \times 4 = 4$             | $4 \times 7 = 28$ |
| <b>Total</b> |                                             | 20 questions                 | <b>60</b>         |

*ESE theory paper — 20 marks total*

**Note:** The ESE theory paper is part of the 60-mark ESE total (the other 40 marks come from the practical/lab examination).

### Practical Test Evaluation Criteria (40 marks)

#### Continuous Evaluation (CIE Practical)

- Preparation & Punctuality — 10
- Experimental Setup & Procedure — 10
- Observation & Data Recording — 5
- Analysis & Result Interpretation — 10
- Viva Voce / Concept Understanding — 10
- Workmanship, Discipline & Teamwork — 5
- **Total: 50** (converted to 10 marks as per scheme)



### Practical Test (ESE Practical)

- Write-up / Procedure — 10
- Experiment Setup & Execution — 10
- Observation & Result / Output — 8
- Viva Voce — 8
- Record Completion & Neatness — 4
- **Total: 40** (directly added to ESE)

#### TEST YOURSELF

### Module-wise Questions with Answers

Practice questions for each module, with answers provided. Use these for self-assessment and exam preparation.

#### Module I — Basics of Computer System

**Q: What is a computer? List its main characteristics.**

**A:** A computer is an electronic device that processes data according to a set of instructions to produce meaningful information. **Characteristics:** Speed, Accuracy, Storage, Versatility, Diligence, and Automation.

**Q: Explain the generations of computers with examples.**

**A: First Generation (Vacuum Tubes):** ENIAC, UNIVAC — large, expensive, high power consumption.

**Second Generation (Transistors):** IBM 1401 — smaller, more reliable.

**Third Generation (ICs):** IBM 360 — multiple transistors on one chip.

**Fourth Generation (Microprocessors):** IBM PC, Apple Mac — personal computers.

**Fifth Generation (AI):** AI systems, quantum computing — parallel processing and intelligence.

**Q: What is the difference between RAM and ROM?**

**A: RAM (Random Access Memory):** Volatile — loses data when power is off. Used for running programs. Can be written to.

**ROM (Read Only Memory):** Non-volatile — retains data without power. Contains firmware (BIOS). Typically read-only (though some types can be written with special procedures).

**Q: Define Algorithm and Flowchart. Draw a flowchart to find the sum of two numbers.**

**A: Algorithm:** A step-by-step procedure to solve a problem.

**Flowchart:** A graphical representation of an algorithm.

**Flowchart for sum of two numbers:** Start → Read A, B → SUM = A + B → Print SUM → Stop. (Refer to the diagram in Module I for visual representation.)

## Module II — Operating Systems

**Q: What is an Operating System? List its main functions.**

**A: Operating System:** System software that acts as an intermediary between hardware and the user. **Functions:** Process Management, Memory Management, File System Management, Device Management, Security & Protection, User Interface, Resource Allocation.

**Q: Explain the types of operating systems based on processing technique.**

**A: Batch Processing OS:** Jobs are processed in batches with no user interaction.

**Time-Sharing OS:** Multiple users share CPU time in small time slices (interactive).

**Real-Time OS:** Responds to events in a guaranteed time frame (used in embedded systems).

**Q: What is the difference between multiprogramming, multitasking, and multiprocessing?**

**A: Multiprogramming:** Multiple programs are loaded into memory, and the CPU switches between them to stay busy.

**Multitasking:** Similar but includes user interaction — multiple tasks appear to run simultaneously.

**Multiprocessing:** Uses multiple CPUs to execute multiple processes simultaneously (requires special hardware).

**Q: Compare Android and iOS as mobile operating systems.**

**A: Android:** Open-source (based on Linux), highly customisable, large app ecosystem, supports multiple vendors.

**iOS:** Closed-source (Apple), highly secure, curated app store, seamless ecosystem, regular updates.

## Module III — Data Processing & Communication

**Q: What is data processing? List its types.**

**A: Data Processing:** The process of collecting, manipulating, and converting raw data into meaningful information. **Types:** Batch Processing, Real-Time Processing, Online

Processing, Distributed Processing.

**Q: Explain the elements of a data communication system.**

**A: Sender (Source):** Device that sends the data.

**Receiver (Destination):** Device that receives the data.

**Message:** The data being transmitted.

**Transmission Medium:** The physical path (wired or wireless).

**Protocol:** A set of rules governing the communication.

**Q: What are the modes of data communication? Give examples.**

**A: Simplex:** One-way communication — Radio broadcast, TV transmission.

**Half-Duplex:** Two-way but only one at a time — Walkie-talkie.

**Full-Duplex:** Two-way simultaneously — Telephone, video conferencing.

**Q: Compare guided and unguided transmission media with examples.**

**A: Guided (Wired):** Twisted Pair (UTP/STP), Coaxial Cable, Optical Fibre. Provide a physical path, higher bandwidth, more secure.

**Unguided (Wireless):** Radio waves, Microwaves, Infrared. No physical path, less secure, more flexible.

## Module IV — Computer Networks & Topologies

**Q: What is a computer network? List its advantages.**

**A: Computer Network:** A set of interconnected computers and devices that communicate with each other. **Advantages:** Resource Sharing, Communication, Centralised Data Management, Cost Efficiency, Reliability, Scalability, Remote Access, Security.

**Q: Explain the types of networks: PAN, LAN, MAN, WAN.**

**A: PAN:** Personal Area Network — few meters (Bluetooth).

**LAN:** Local Area Network — building/campus (Ethernet, WiFi).

**MAN:** Metropolitan Area Network — city (cable TV network).

**WAN:** Wide Area Network — country/global (the Internet).

**Q: Compare star, ring, bus, and mesh topologies.**

**A: Star:** All devices connected to a central hub — easy to manage, single point of failure.

**Ring:** Each device connected to two others in a ring — predictable, one break breaks the network.

**Bus:** Single central cable — simple, low cost, single point of failure.

**Mesh:** Every device connected to every other — highly redundant, robust, expensive.

**Q: What is the difference between a hub, a switch, and a router?**

**A: Hub:** Broadcasts data to all devices (Layer 1 — Physical). Inefficient.

**Switch:** Forwards data to the specific device using MAC address (Layer 2 — Data Link). Efficient.

**Router:** Connects different networks using IP address (Layer 3 — Network). Routes data between networks.

**PRACTICE EXAM**

## Model Question Paper

A practice paper based on the official pattern. **Note:** This is a practice model prepared for study purposes, not an official SITTTR question paper.

**Course:** Computer Applications in Management (1254) | **Semester:** I | **Time:** 2.5 hours | **Max Marks:** 60

Part A:  $8 \times 1 = 8$  marks | Part B:  $8 \times 3 = 24$  marks | Part C:  $4 \times 7 = 28$  marks

### Part A — Very Short Answer (1 mark each)

*Answer any 8 questions. Each question carries 1 mark.*

1. What is a computer?
2. Define RAM and ROM.
3. What is an Operating System?
4. List any two functions of an OS.
5. What is data processing?
6. What is a simplex communication mode?
7. What is a LAN?
8. Name any two network topologies.
9. What is a router?
10. What is an IP address?

### Part B — Short Answer (3 marks each)

*Answer any 8 questions. Each question carries 3 marks.*

1. Explain the characteristics of computers.
2. Explain the generations of computers.
3. Explain the layers of an operating system.
4. What is the difference between multiprogramming and multitasking?
5. Explain the elements of a data communication system.

6. What are the modes of data communication? Give examples.
7. Explain the types of networks: PAN, LAN, MAN, WAN.
8. Compare star and ring topologies.
9. What is the difference between a hub and a switch?
10. Explain the basics of the Internet (ISP, web server, browser).

### Part C — Long Answer (7 marks each)

Answer any 4 questions. Each question carries 7 marks. (One set with choice per module.)

1. **Module I:** Explain the components of a computer system with a block diagram. OR Explain the types of computer memory with a hierarchy diagram.
2. **Module II:** Explain the functions of an operating system in detail. OR Compare Windows and Linux operating systems.
3. **Module III:** Explain the types of data transmission media (guided and unguided) with examples and diagrams. OR Explain the elements of a data communication system and the modes of data communication.
4. **Module IV:** Explain network topologies (bus, ring, star, mesh) with diagrams, advantages, and disadvantages. OR Explain network components (hub, switch, router, gateway) and their functions.

 **Practice tip:** Attempt this paper under timed conditions. Use the Q&A section for reference. Check your answers against the module content.

#### LAST-MINUTE REVIEW

### Quick Revision

Condensed key points from each module, common mistakes, and an exam checklist.

#### Module I — Computer Basics

- **Characteristics:** Speed, Accuracy, Storage, Versatility, Diligence, Automation.
- **Generations:** Vacuum tubes → Transistors → ICs → Microprocessors → AI.
- **Memory:** RAM (volatile), ROM (non-volatile), Cache, HDD, SSD.
- **Software:** System (OS, drivers) vs Application (MS Office, browsers).
- **Languages:** Machine, Assembly, High-Level (C, Python).
- **Translators:** Compiler (whole program) vs Interpreter (line by line).

#### Module II — Operating Systems

- **Functions:** Process, Memory, File, Device, Security, UI, Resource.
- **Types:** Batch, Time-Sharing, Real-Time.
- **Techniques:** Multiprogramming, Multitasking, Multiprocessing.
- **Popular OS:** Windows (GUI, proprietary), Linux (open-source, stable).

- **Mobile OS:** Android (open), iOS (closed), features: touch, apps, cloud.

### 📌 Module III — Data Processing & Communication

- **Data Processing:** Batch, Real-Time, Online, Distributed.
- **File Types:** Master, Transaction, Reference, Report.
- **DBMS:** Reduces redundancy, ensures consistency, uses SQL.
- **Communication Modes:** Simplex (one-way), Half-Duplex (alternate), Full-Duplex (simultaneous).
- **Transmission Media:** Guided (Twisted Pair, Coaxial, Fibre) vs Unguided (Radio, Microwave, Infrared).
- **Modem:** Modulates (digital → analog) and Demodulates (analog → digital).

### 📌 Module IV — Computer Networks

- **Types:** PAN (few meters), LAN (building), MAN (city), WAN (global).
- **Topologies:** Bus (single cable), Ring (circle), Star (central hub), Mesh (fully connected).
- **Components:** Hub (broadcast), Switch (MAC address), Router (IP address), Gateway (protocol conversion).
- **Internet:** ISP, Web Server, Browser, IP Address, Domain Name, Intranet.
- **Commands:** ipconfig/ifconfig, ping, tracert/traceroute, nslookup.
- **Cloud:** Amazon S3, EC2 — storage and computing in the cloud.

### ⚠️ Common Mistakes to Avoid

**Confusing RAM and ROM:** RAM is volatile, ROM is non-volatile. Remember: RAM can be written to; ROM is typically read-only.

**Mixing up topologies:** Star has a central hub; Ring forms a circle; Bus has a single cable; Mesh has multiple connections.

**Hub vs Switch vs Router:** Hub broadcasts to all, Switch uses MAC addresses, Router uses IP addresses.

**Compiler vs Interpreter:** Compiler translates the whole program; Interpreter translates line by line.

### ✅ Exam Checklist

- ☐ I know the characteristics and generations of computers.
- ☐ I can explain the components of a computer system and the memory hierarchy.
- ☐ I understand the functions and types of operating systems.

- ☐ I can differentiate between multiprocessing, multitasking, and multiprocessor.
- ☐ I know the types of data processing and file management.
- ☐ I understand the elements of data communication and transmission media.
- ☐ I can explain network topologies and compare them.
- ☐ I know the functions of hub, switch, router, and gateway.
- ☐ I understand Internet basics (ISP, web server, browser, IP, domain).
- ☐ I can use ipconfig/ifconfig, ping, and traceroute commands.
- ☐ I have practiced the model paper and reviewed the Q&A.

## KEY TERMS

# Glossary

Important technical terms and their one-line meanings.

| Term       | Meaning                                                                      |
|------------|------------------------------------------------------------------------------|
| CPU        | Central Processing Unit — the "brain" of the computer                        |
| RAM        | Random Access Memory — volatile memory for running programs                  |
| ROM        | Read Only Memory — non-volatile memory for firmware                          |
| HDD        | Hard Disk Drive — magnetic storage for permanent data                        |
| SSD        | Solid State Drive — flash-based storage, faster than HDD                     |
| OS         | Operating System — manages hardware and provides a platform for applications |
| GUI        | Graphical User Interface — visual way to interact with a computer            |
| CLI        | Command Line Interface — text-based way to interact with a computer          |
| DBMS       | Database Management System — software for managing structured data           |
| SQL        | Structured Query Language — language for querying databases                  |
| LAN        | Local Area Network — network within a limited area                           |
| WAN        | Wide Area Network — network covering a large geographic area                 |
| PAN        | Personal Area Network — network for personal devices                         |
| MAN        | Metropolitan Area Network — network covering a city                          |
| IP Address | Internet Protocol Address — unique identifier for a device on a network      |
| DNS        | Domain Name System — translates domain names to IP addresses                 |
| ISP        | Internet Service Provider — company that provides internet access            |
| HTTP       | HyperText Transfer Protocol — protocol for web communication                 |
| HTTPS      | HTTP Secure — secure version of HTTP                                         |
| FOSS       | Free and Open Source Software — software with freely available source code   |

| Term | Meaning                                                                              |
|------|--------------------------------------------------------------------------------------|
| BIOS | Basic Input Output System — firmware that starts the computer                        |
| SMPS | Switched Mode Power Supply — power supply for computers                              |
| CMOS | Complementary Metal-Oxide-Semiconductor — battery-powered memory for system settings |

*Glossary of important terms*

#### FURTHER READING

## References & Resources

Textbooks, reference books, and web resources listed in the official syllabus.

### Textbooks

1. Rajaraman — *Fundamentals of Computers*, Prentice Hall of India, 2003.
2. P.K Sinha, Priti Sinha — *Computer Fundamentals*, Kalyani Publishers, 2nd Edition.
3. Shelly — *Business Data Communication*, Thomson Learning, Bombay, 2003.
4. White — *Data Communication & Computer Network*, Thomson Learning, Bombay, 2012.

### Reference Books

1. B. Ram — *Computer Fundamentals*, New Age International, 2000.
2. Behrouz A. Forouzan — *Data Communications and Networking*, McGraw Hill Education, 2017.

### Web Resources (from syllabus)

- Fortinet — Wireless Network (Module III)
- Lenovo — What is Modem (Module III)
- Network Cable Crimping and Testing Tools (Module IV)